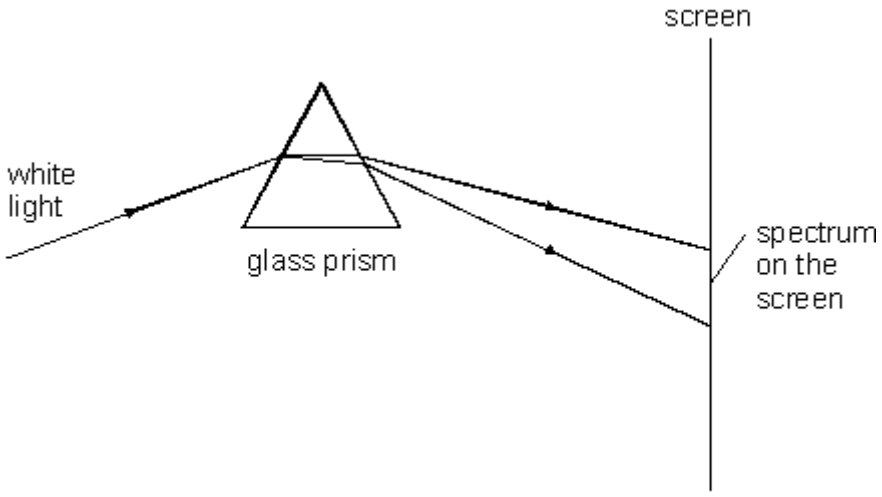
Q1.

When white light is shone through a glass prism the light bends and splits into the colours of the spectrum.



(a) (i) What word describes the bending of light as it enters and leaves glass?

……..……

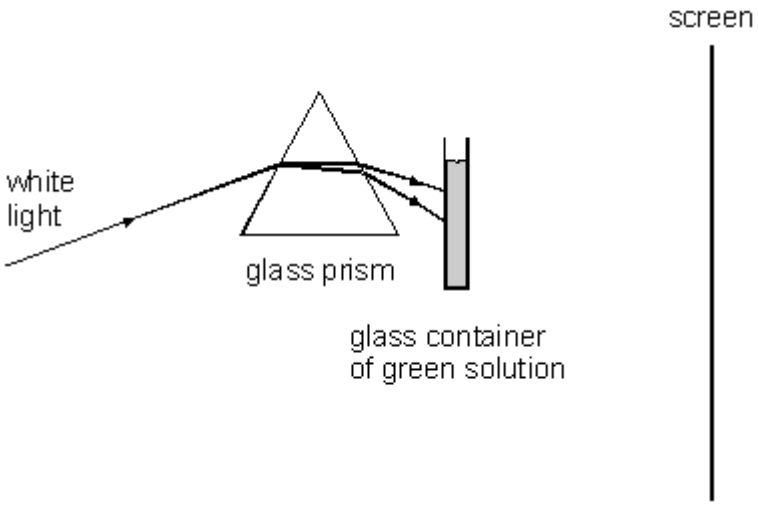
1 mark

(ii) What word describes the splitting of light into the colours of the spectrum?

……..……

1 mark

(b) Some leaves from a buttercup plant were ground up in a solvent and filtered to give a green solution of chlorophyll. A glass container of this green solution was put in the rays of coloured light.



What change in the spectrum would you see on the screen?
Explain your answer.

……

……

……

2 marks

(c) Why is it necessary to grind up the buttercup leaves to release the chlorophyll from the cells?

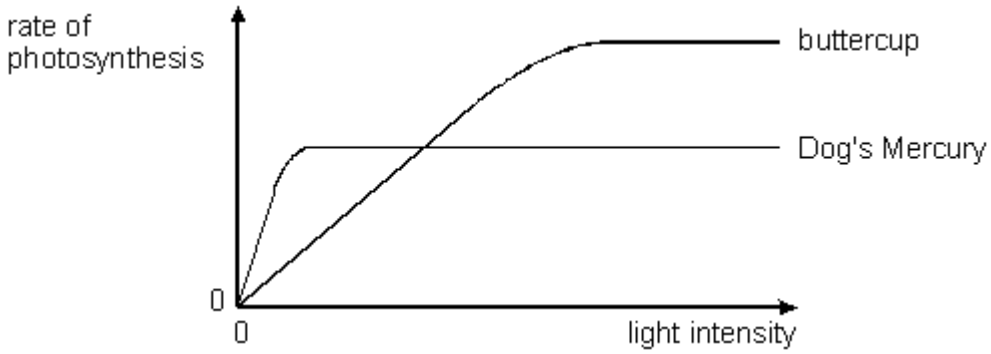
……

……

……

1 mark

(d) Buttercup plants grow mainly in open fields. Dog's Mercury is a plant which grows mainly in woodland. The graph shows how the rate of photosynthesis in these two plants changes as the light intensity changes.



Why do Dog's Mercury plants grow better than buttercups in woodland?
Use the graph to help you.

……

……

……

1 mark
Maximum 6 marks

Q1.

(a) (i) refraction

1 (L6)

(ii) dispersion

1 (L7)

(b) **One mark is for what would be seen on the screen.**
The second mark is for the explanation.

- only the green part would be seen
  accept 'only the green'

1 (L7)

- other colours are absorbed **or** removed by the green solution
  accept 'only green can go through'

1 (L7)

(c) any **one** from

- to break down the cell walls
- to break open the cells
- because the green substance is inside the cells **or** chloroplasts
  accept 'to break up the chloroplasts'
  accept 'to break down the cuticle'
  *do **not** accept 'to break them down'*

1 (L7)

(d) **A comparison between the rate of photosynthesis in Dog's Mercury and buttercup must be made.**

any **one** from

- Dog's Mercury has a higher rate of photosynthesis than buttercups in low light
  accept 'Dog's Mercury has a higher rate of photosynthesis in low light'
  accept 'it can photosynthesise better than buttercups in the shade'
  accept 'Dog's Mercury can make more food in the shade'
  *do **not** accept 'it can photosynthesise in the dark'*
- buttercups have a lower rate of photosynthesis than Dog's Mercury in low light
  accept 'buttercups have a lower rate of photosynthesis in lower light'
- buttercups reach their maximum rate of photosynthesis at **higher** light intensities
- Dog's Mercury reaches its maximum rate of photosynthesis at **lower** light intensities

1 (L7)

[6]